17/23

Blessed G/Michael Catholic Secondary School

1st Semester Chemistry Test One for Grade 10th

Oct, 2018 E.C Time Allowed 1 hr

element	Al	O	H	S	C	Mg
mass	27	16	1	32	12	24

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I. Write true if the statement is correct and write false if the statement is wrong

false 1. Most metallic elements are strong oxidizing agent

false 2. In a chemical reaction actual yield of product is usually less than theoretical yield of the product

false 3. Neutralization and double displacement reaction is an example of redox reaction

false 4. The sum of oxidation number of all atoms in neutral compound is equal to the charge of the ion

I. Choose the best answer among the given alternatives

B 5. Which of the following is not an example of chemical change?

A. rotting of egg by bacteria
B. the formation of cloud
C. Digesting of food
D. rusting of iron

C 6. Which one of the following reaction is synthesis reaction?

A. $CaCO_3 \rightarrow CaO + CO_2$
B. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
C. $2CO + O_2 \rightarrow 2CO_2$
D. $2Na_3PO_4 + 3Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + 6NaOH$

C 7. A type of reaction in which two compounds react to form new compounds by exchange of positive and negative ion is

A. Decomposition reaction
B. Single displacement reaction
C. Double displacement reaction
D. Combination reaction

B 8. How many mole of Al_2O_3 can be formed when a mixture of 2 moles of Al and 0.6 mole of oxygen gas is ignited?

A. 0.2 mole B. 0.4 C. 2 mole D. 0.6 mole

D 9. In which of the following species does nitrogen have the highest oxidation number?

A. N_2O_3 B. N_2H_4 C. N_2O_4 D. N_2O_5

D 10. Given the reaction; $2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H2$. Which one of the following substances is an oxidizing agent?

A. $Al_2(SO_4)_3$. B. H_2. C. Al. D. H_2SO_4

B 11. How many mole of CO_2 are present in 176 g of CO_2?

A. 2 mole B. 4 mole C. 8mole D. 16 mole

B 12. How many atoms and ions are there in $Al_2(SO_4)_3$?

A. 11 atom and 5 ion
B. 17 atom and 5 ion
C. 17 atom and 12 ion
D. 11 atom and 12 ion

B 13. What is the oxidation state of phosphorus in $H_2P_2O_7^{-2}$?

A. -5 B. +5 C. -6 D. +6

B 14. How many litres of water are produced in burning of 20g hydrogen gas at STP?

A. 448.0L B. 224.0L C. 22.4L D. 44.8L.

A 15. How many litres of hydrogen are required to react with 0.5 L carbon monoxide gas according to the following reaction? $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$

A. 1.0L B. 0.2L C. 0.1L D. 0.5L

D A 16. Which the following statement is true about oxidizing and reducing agent

A. A reducing agent accept electron in redox reaction
B. An oxidizing agent lose electron in redox reaction
C. A reducing agent contains an element that undergoes reduction
D. An oxidizing agent causes reduction during redox reaction

A B 17. How many mole of sodium oxide are needed to react with water to produce 120g of sodium hydroxide? [Mass of Na= 23]

A. 0.5 mole B. 1.5 mole C. 0.15 mole D. 0.2 mole

B 18. What mas of magnesium oxide are produced when burning of 4g magnesium metal?

$2Mg + O_2 \rightarrow 2MgO$

A. 6g B. 6.6g C. 0.6g D. 8g

C 19. For the following reaction which one is not redox reaction?

A. $2Na + Cl_2 \rightarrow 2NaCl$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $NaOH + HCl \rightarrow NaCl + H_2O$
D. $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$.

A D 20. If 5g of sulphur dioxide is actually formed from the reaction of 10g of oxygen with sufficient sulphur, what is the percentage yield of sulphur dioxide?

A. 50% B. 75%. C. 90%. D.25%

Give a short answer

21. Write the balanced chemical equation for the following reaction;

A. iron (III) sulphate reacts with potassium hydroxide to forms potassium sulphate and iron oxide

$2Fe_2(SO_4)_3 + 6KOH \rightarrow 3K_2SO_4 + 2Fe_2O_3$

B. $H_2SO_4 + NaOH \rightarrow NaSO_4 + H_2O$

$3H_2SO_4 + 3NaOH \rightarrow 3NaSO_4 + 3H_2O$

22. 20 g of $CaCO_3$ and 30 g of HCl are mixed, according to the following reaction;

(mass,Ca=40, Cl=35, O=16, C=12)

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

A. Which substance is limiting reactant CaCO3 ?

B. How many grams of H_2O would be formed 3.6 ?

C. How many grams of the reactant remain unreacted 15.6 ?

HCl is excess so, α $\frac{2HCl}{72g} \times \frac{3.6}{18} H_2O$ $\alpha = \frac{3.6(72)}{18} = 14.4g$

then 30g - 14.4 = 15.6g of HCl is unreacted

$\frac{10}{32} = \frac{x}{64}$ $\frac{640}{32} = \frac{32x}{32}$ $x = 20$

$S + O_2 \rightarrow SO_2$

32 64

% yield: $\frac{Act}{the} \times 100$

$\frac{500}{20} = 25$